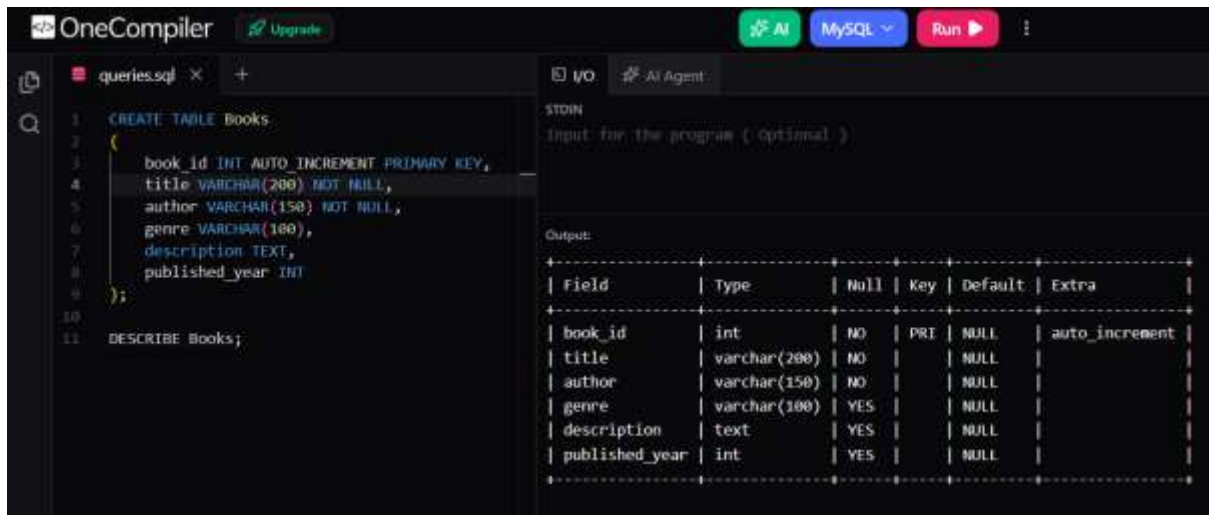


WEEK 4 — BOOKFLOW INTELLIGENT CATALOG & API BLUEPRINT

Q01 — CREATE BOOKS TABLE

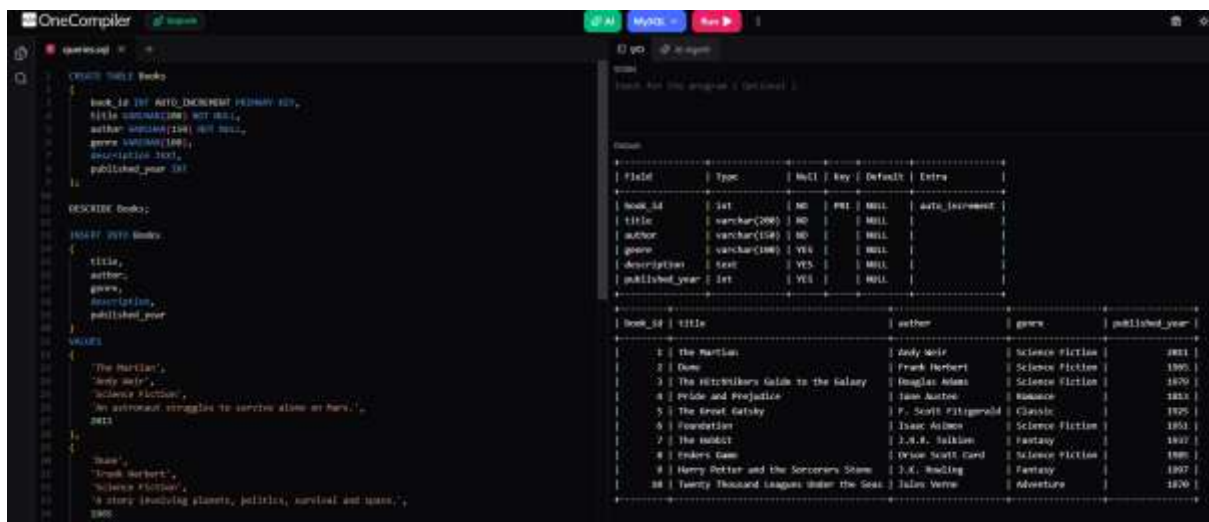


```
1 CREATE TABLE Books
2 (
3     book_id INT AUTO INCREMENT PRIMARY KEY,
4     title VARCHAR(200) NOT NULL,
5     author VARCHAR(150) NOT NULL,
6     genre VARCHAR(100),
7     description TEXT,
8     published_year INT
9 );
10
11 DESCRIBE Books;
```

Output:

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	auto_increment
title	varchar(200)	NO		NULL	
author	varchar(150)	NO		NULL	
genre	varchar(100)	YES		NULL	
description	text	YES		NULL	
published_year	int	YES		NULL	

Q02 — INSERT BOOK DATA



```
1 CREATE TABLE Books
2 (
3     book_id INT AUTO INCREMENT PRIMARY KEY,
4     title VARCHAR(200) NOT NULL,
5     author VARCHAR(150) NOT NULL,
6     genre VARCHAR(100),
7     description TEXT,
8     published_year INT
9 );
10
11 DESCRIBE Books;
12
13 INSERT INTO Books
14 (
15     title,
16     author,
17     genre,
18     description,
19     published_year
20 )
21 VALUES
22 (
23     'The Martian',
24     'Andy Weir',
25     'Science Fiction',
26     'An astronaut struggles to survive alone on Mars.',
27     2011
28 ),
29 (
30     'Dune',
31     'Frank Herbert',
32     'Science Fiction',
33     'A story involving planets, politics, survival and space.',
34     1965
35 ),
36 (
37     'The Hitchhiker's Guide to the Galaxy',
38     'Douglas Adams',
39     'Science Fiction',
40     'A story involving planets, politics, survival and space.',
41     1979
42 ),
43 (
44     'Pride and Prejudice',
45     'Jane Austen',
46     'Romance',
47     'A story involving planets, politics, survival and space.',
48     1813
49 ),
50 (
51     'The Great Gatsby',
52     'F. Scott Fitzgerald',
53     'Classic',
54     'A story involving planets, politics, survival and space.',
55     1925
56 ),
57 (
58     'Foundation',
59     'Isaac Asimov',
60     'Science Fiction',
61     'A story involving planets, politics, survival and space.',
62     1951
63 ),
64 (
65     'The Hobbit',
66     'J.R.R. Tolkien',
67     'Fantasy',
68     'A story involving planets, politics, survival and space.',
69     1937
70 ),
71 (
72     'Ender's Game',
73     'Orson Scott Card',
74     'Science Fiction',
75     'A story involving planets, politics, survival and space.',
76     1985
77 ),
78 (
79     'Harry Potter and the Sorcerer's Stone',
80     'J.K. Rowling',
81     'Fantasy',
82     'A story involving planets, politics, survival and space.',
83     1997
84 ),
85 (
86     'Twenty Thousand Leagues Under the Sea',
87     'Julius Verne',
88     'Adventure',
89     'A story involving planets, politics, survival and space.',
90     1870
91 );
```

Output:

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	auto_increment
title	varchar(200)	NO		NULL	
author	varchar(150)	NO		NULL	
genre	varchar(100)	YES		NULL	
description	text	YES		NULL	
published_year	int	YES		NULL	

book_id	title	author	genre	published_year
1	The Martian	Andy Weir	Science Fiction	2011
2	Dune	Frank Herbert	Science Fiction	1965
3	The Hitchhiker's Guide to the Galaxy	Douglas Adams	Science Fiction	1979
4	Pride and Prejudice	Jane Austen	Romance	1813
5	The Great Gatsby	F. Scott Fitzgerald	Classic	1925
6	Foundation	Isaac Asimov	Science Fiction	1951
7	The Hobbit	J.R.R. Tolkien	Fantasy	1937
8	Ender's Game	Orson Scott Card	Science Fiction	1985
9	Harry Potter and the Sorcerer's Stone	J.K. Rowling	Fantasy	1997
10	Twenty Thousand Leagues Under the Sea	Julius Verne	Adventure	1870

```

),
(
    'Pride and Prejudice',
    'Jane Austen',
    'Romance',
    'A classic story about relationships and social expectations.',
    1813
),
(
    'The Great Gatsby',
    'F. Scott Fitzgerald',
    'Classic',
    'A story of ambition, wealth, love and the American dream.',
    1925
),
(
    'Foundation',
    'Isaac Asimov',
    'Science Fiction',
    'A futuristic story about mathematics, civilization and space.',
    1951
),
(
    'The Hobbit',
    'J.R.R. Tolkien',
    'Fantasy',
    'A fantasy adventure involving Bilbo Baggins and a dangerous journey.',
    1937
),
(
    'Ender's Game',
    'Orson Scott Card',
    'Science Fiction',
    'A young student is trained for an interstellar conflict.',
    1985
),
(
    'Harry Potter and the Sorcerer's Stone',
    'J.K. Rowling',
    'Fantasy',
    'A young wizard begins his magical education.',
    1997
),

```

```

(
    'Twenty Thousand Leagues Under the Seas',
    'Jules Verne',
    'Adventure',
    'A fantastic underwater adventure aboard a mysterious submarine.'
    1870
);

```

```

SELECT
    book_id,
    title,
    author,
    genre,
    published_year
FROM Books
ORDER BY book_id;

```

Q03 — ADD EMBEDDING COLUMN

```
117 SELECT
118   book_id,
119   title,
120   author,
121   genre,
122   published_year
123 FROM Books
124 ORDER BY book_id;
125
126 ALTER TABLE Books
127 ADD COLUMN embedding TEXT;
128
129 DESCRIBE Books;
```

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	auto_increment
title	varchar(200)	NO		NULL	
author	varchar(150)	NO		NULL	
genre	varchar(100)	YES		NULL	
description	text	YES		NULL	
published_year	int	YES		NULL	
embedding	json	YES		NULL	

Q04 — INSERT MOCK VECTORS

```
129 DESCRIBE Books;
130
131 INSERT INTO Books (book_id, title, author, genre, published_year, embedding)
132 VALUES (1, 'The Martian', 'Andy Weir', 'Science Fiction', 2011, '[0.90, 0.90, 0.20]');
133
134 INSERT INTO Books (book_id, title, author, genre, published_year, embedding)
135 VALUES (2, 'Dune', 'Frank Herbert', 'Science Fiction', 1965, '[0.45, 0.75, 0.25]');
136
137 INSERT INTO Books (book_id, title, author, genre, published_year, embedding)
138 VALUES (3, 'The Hitchhiker's Guide to the Galaxy', 'Douglas Adams', 'Science Fiction', 1979, '[0.90, 0.70, 0.30]');
139
140 INSERT INTO Books (book_id, title, author, genre, published_year, embedding)
141 VALUES (4, 'Pride and Prejudice', 'Jane Austen', 'Romance', 1813, '[0.10, 0.20, 0.90]');
142
143 INSERT INTO Books (book_id, title, author, genre, published_year, embedding)
144 VALUES (5, 'The Great Gatsby', 'F. Scott Fitzgerald', 'Classic', 1925, '[0.15, 0.30, 0.80]');
145
146 INSERT INTO Books (book_id, title, author, genre, published_year, embedding)
147 VALUES (6, 'Foundation', 'Isaac Asimov', 'Science Fiction', 1951, '[0.95, 0.05, 0.20]');
148
149 INSERT INTO Books (book_id, title, author, genre, published_year, embedding)
150 VALUES (7, 'The Hobbit', 'J.R.R. Tolkien', 'Fantasy', 1937, '[0.30, 0.75, 0.20]');
151
152 INSERT INTO Books (book_id, title, author, genre, published_year, embedding)
153 VALUES (8, 'Enders Game', 'Orson Scott Card', 'Science Fiction', 1985, '[0.85, 0.85, 0.15]');
154
155 INSERT INTO Books (book_id, title, author, genre, published_year, embedding)
156 VALUES (9, 'Harry Potter and the Sorcerer's Stone', 'J.K. Rowling', 'Fantasy', 1997, '[0.20, 0.55, 0.95]');
157
158 INSERT INTO Books (book_id, title, author, genre, published_year, embedding)
159 VALUES (10, 'Twenty Thousand Leagues Under the Sea', 'Jules Verne', 'Adventure', 1870, '[0.70, 0.90, 0.40]');
```

book_id	title	author	genre	published_year
1	The Martian	Andy Weir	Science Fiction	2011
2	Dune	Frank Herbert	Science Fiction	1965
3	The Hitchhiker's Guide to the Galaxy	Douglas Adams	Science Fiction	1979
4	Pride and Prejudice	Jane Austen	Romance	1813
5	The Great Gatsby	F. Scott Fitzgerald	Classic	1925
6	Foundation	Isaac Asimov	Science Fiction	1951
7	The Hobbit	J.R.R. Tolkien	Fantasy	1937
8	Enders Game	Orson Scott Card	Science Fiction	1985
9	Harry Potter and the Sorcerer's Stone	J.K. Rowling	Fantasy	1997
10	Twenty Thousand Leagues Under the Sea	Jules Verne	Adventure	1870

Q05 — VERIFY ALL VECTORS

```
160 SET embedding = JSON_ARRAY(0.90, 0.90, 0.20);
161 WHERE book_id = 1;
162
163 UPDATE Books
164 SET embedding = JSON_ARRAY(0.20, 0.55, 0.95)
165 WHERE book_id = 9;
166
167 UPDATE Books
168 SET embedding = JSON_ARRAY(0.70, 0.90, 0.40)
169 WHERE book_id = 10;
170
171 SELECT
172   book_id,
173   title,
174   embedding
175 FROM Books
176 ORDER BY book_id;
```

book_id	title	embedding
1	The Martian	[0.90, 0.90, 0.20]
2	Dune	[0.45, 0.75, 0.25]
3	The Hitchhiker's Guide to the Galaxy	[0.90, 0.70, 0.30]
4	Pride and Prejudice	[0.10, 0.20, 0.90]
5	The Great Gatsby	[0.15, 0.30, 0.80]
6	Foundation	[0.95, 0.05, 0.20]
7	The Hobbit	[0.30, 0.75, 0.20]
8	Enders Game	[0.85, 0.85, 0.15]
9	Harry Potter and the Sorcerer's Stone	[0.20, 0.55, 0.95]
10	Twenty Thousand Leagues Under the Sea	[0.70, 0.90, 0.40]

Q06 — L2 DISTANCE SEARCH

```

254 SELECT
255     book_id,
256     title,
257     author,
258     genre,
259     ROUND(
260         SQRT(
261             POW(
262                 CAST(
263                     JSON_EXTRACT(embedding, '$[0]')
264                     AS DECIMAL(10,6)
265                 ) - 0.90,
266                 2
267             )
268             +
269             POW(
270                 CAST(
271                     JSON_EXTRACT(embedding, '$[1]')
272                     AS DECIMAL(10,6)
273                 ) - 0.80,
274                 2
275             )
276             +
277             POW(
278                 CAST(
279                     JSON_EXTRACT(embedding, '$[2]')
280                     AS DECIMAL(10,6)
281                 ) - 0.70,
282                 2
283             )
284         ),
285         4
286     ) AS l2_distance
287 FROM Books
288 WHERE embedding IS NOT NULL
289 ORDER BY l2_distance ASC
290 LIMIT 3;

```

Field	Type	Null	Key	Default	Extra
book_id	Int	NO	PK	NULL	auto_increment
title	varchar(200)	NO		NULL	
author	varchar(150)	NO		NULL	
genre	varchar(100)	YES		NULL	
description	text	YES		NULL	
published_year	Int	YES		NULL	
embedding	json	YES		NULL	

book_id	title	embedding
1	The Martian	[0.90, 0.80, 0.70]
2	Dune	[0.85, 0.75, 0.25]
3	The Hitchhiker's Guide to the Galaxy	[0.80, 0.70, 0.30]
4	Pride and Prejudice	[0.10, 0.20, 0.90]
5	The Great Gatsby	[0.15, 0.30, 0.80]
6	Foundation	[0.95, 0.05, 0.20]
7	The Hobbit	[0.30, 0.75, 0.90]
8	Enders Game	[0.80, 0.85, 0.15]
9	Harry Potter and the Sorcerers Stone	[0.20, 0.55, 0.95]
10	Twenty Thousand Leagues Under the Seas	[0.70, 0.90, 0.40]

book_id	title	author	genre	l2_distance
1	The Martian	Andy Weir	Science Fiction	0
8	Enders Game	Orson Scott Card	Science Fiction	0.0725
2	Dune	Frank Herbert	Science Fiction	0.0906

Q07 — COSINE SIMILARITY SEARCH

```

291 SELECT
292     ROUND(
293         SQRT(
294             POW(
295                 CAST(
296                     JSON_EXTRACT(embedding, '$[0]')
297                     AS DECIMAL(10,6)
298                 ) - 0.90,
299                 2
300             )
301             +
302             POW(
303                 CAST(
304                     JSON_EXTRACT(embedding, '$[1]')
305                     AS DECIMAL(10,6)
306                 ) - 0.80,
307                 2
308             )
309             +
310             POW(
311                 CAST(
312                     JSON_EXTRACT(embedding, '$[2]')
313                     AS DECIMAL(10,6)
314                 ) - 0.70,
315                 2
316             )
317         ),
318         4
319     ) AS cosine_similarity
320 FROM Books
321 WHERE embedding IS NOT NULL
322 ORDER BY cosine_similarity DESC
323 LIMIT 3;

```

Field	Type	Null	Key	Default	Extra
author	varchar(150)	NO		NULL	
genre	varchar(100)	YES		NULL	
description	text	YES		NULL	
published_year	Int	YES		NULL	
embedding	json	YES		NULL	

book_id	title	embedding
1	The Martian	[0.90, 0.80, 0.70]
2	Dune	[0.85, 0.75, 0.25]
3	The Hitchhiker's Guide to the Galaxy	[0.80, 0.70, 0.30]
4	Pride and Prejudice	[0.10, 0.20, 0.90]
5	The Great Gatsby	[0.15, 0.30, 0.80]
6	Foundation	[0.95, 0.05, 0.20]
7	The Hobbit	[0.30, 0.75, 0.90]
8	Enders Game	[0.80, 0.85, 0.15]
9	Harry Potter and the Sorcerers Stone	[0.20, 0.55, 0.95]
10	Twenty Thousand Leagues Under the Seas	[0.70, 0.90, 0.40]

book_id	title	author	genre	l2_distance
1	The Martian	Andy Weir	Science Fiction	0
8	Enders Game	Orson Scott Card	Science Fiction	0.0725
2	Dune	Frank Herbert	Science Fiction	0.0906

book_id	title	author	genre	cosine_similarity
1	The Martian	Andy Weir	Science Fiction	1
2	Dune	Frank Herbert	Science Fiction	0.9996
8	Enders Game	Orson Scott Card	Science Fiction	0.9983